

C Programming: Pointers, Memory Layout & Dynamic Allocation

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1. Pointer Fundamentals & Architecture

A pointer in C contains the direct hardware hexadecimal address of another variable.
Dereferencing with * accesses value; the address-of operator & retrieves memory location.
Memory hierarchy consists of Text/Code, Initialized Data (BSS), Stack, and Dynamic Heap.

2. Dynamic Allocation (malloc / calloc / realloc / free)

Heap memory is allocated at runtime using malloc(size_in_bytes) and returns (void*).
calloc allocates contiguous elements and zeros memory automatically.
Every allocated block MUST be released with free() to prevent memory leaks and fragmentation.

3. Pointer Arithmetic & Array Equivalence

In C, array subscript notation a[i] is semantically identical to pointer arithmetic *(a + i).
Pointer addition increments by the size of the underlying data type (sizeof(T)).